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Some factors influencing the effects of temporary mother–infant separation: some experiments with rhesus monkeys

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SYNOPSIS Some experiments, reported in detail elsewhere, on the effects of mother–infant separation in rhesus monkeys are here reviewed and compared. They involved 4 groups – one in which mothers were removed for 13 days leaving the infant in the social group; one in which infants were removed; one in which mothers and infants were removed and separated; and one in which mothers and infants were removed but not separated. The nature of the separation experience had a profound effect on the infant's response: infants left in a familiar environment while their mothers were removed showed marked but brief 'protest' and then profound 'despair', whilst infants removed to a strange cage showed more prolonged 'protest'. A major factor determining the effects of the separation experience in the weeks following reunion is the degree to which the mother–infant relationship has been disturbed by it. The multiplicity of factors affecting the outcome of a separation experience are discussed.

INTRODUCTION

When separated from their mothers, young children are likely to show acute distress followed by a slow process of adaptation: the successive stages are often referred to as involving Protest, Despair and Detachment (Bowlby, 1960, 1969; Robertson, 1953, 1970; Spitz, 1950). Furthermore, a variety of types of evidence indicate that such a separation experience can have long-term effects, though these are usually detectable only in certain circumstances (e.g. Rutter, 1971, 1972; Douglas, 1975). There is, however, disagreement about which aspects of a separation experience are responsible for producing the observed effects – is it absence of the mother (Bowlby, 1960), deprivation of emotional support (Howells, 1970), associated stresses (Robertson & Robertson, 1971), or some other factor or factors? Since it is inevitable that separations should occur, it is important to know which aspects of a separation experience produce the effects observed, which children are most at risk, and how ill-effects can be ameliorated.

Previous work (reviewed Hinde & Spencer-Booth, 1971, and refs. cited; Kaufman & Rosenblum, 1967; Suomi, 1974; Kaufman, 1974) has shown that during a separation experience infant monkeys show changes in behaviour comparable to those described for human children. Furthermore, separation can have long-term effects (Spencer-Booth & Hinde, 1971*a*). Perhaps, therefore, studies of monkeys can suggest answers to the questions posed above – answers whose applicability to the human case would be worth while assessing. There is no implication here that monkey data are directly applicable to man, only that their relative simplicity sometimes permits the isolation of principles which may subsequently prove also to be valid for man (Hinde, 1976).

Studies from this laboratory have involved monkeys living in small social groups. Whilst this complicates their interpretation, animals living in near-isolation in small laboratory cages

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often show neurotic symptoms, and this produces problems of both interpretation and generality. We prefer the compromise involved in a moderately complex social situation permitting moderately precise data collection and control.

This paper reviews 4 experiments attempting to assess the relative importance of 3 factors which could contribute to the effects of a separation experience on the infant's behaviour in the weeks immediately following reunion with the mother. These were

(*a*) separation from the mother *per se*, with the associated deprivation of maternal care;

(*b*) continued presence of the infant in the home environment during the separation period, with the possibility of adverse conditioning to that environment;

(*c*) disturbance to the mother–infant relationship.

The first experiment involved removing mothers from the home pen for periods of 13 days (Mother-Removed, or M/R procedure; Spencer-Booth & Hinde, 1971*b*). We then studied a group in which the infants were removed for 13 days, the mother remaining in the home pen (Infant-Removed, or I/R group; Hinde & Davies, 1972*a*, *b*). This suggested both that the circumstances in which the infant was kept during separation affected its response, and that the mother's interactions with social companions on return to the pen could limit her attentiveness to the baby and thus affect its recovery. Two further experiments involved removal of infant and mother and separating them from each other for 13 days before reuniting them in the home pen (Mother and Infant Removed and Separated, or M–I/R group), and removal of infant and mother together, without separating them from each other ((MI)/R group, McGinnis, 1975, in preparation). This paper is concerned primarily with a comparison of the 4 experiments. The findings are discussed also in relation to those from other experiments with non-human primates and from studies of children.

MATERIALS AND METHODS

GENERAL

Details of subjects and methods are given in the other papers cited. Only essential issues are summarized here.

TABLE 1

THE EXPERIMENTAL GROUPS

Group	*N*	Pen during separation: Mother	Pen during separation: Infant	Separation	Reference
M/R	6	Strange	Home	Yes	Spencer-Booth & Hinde, 1971*b*
I/R	6	Home	Strange	Yes	Hinde & Davies, 1972*a*, *b*
M–I/R	7	Strange	Strange	Yes	McGinnis, 1975, in prep.
(MI)/R	7	Strange	Strange	No	McGinnis, 1975, in prep.

SUBJECTS

Each social group consisted of an adult male rhesus monkey, 2–4 adult females and their young. The infants had not previously been separated from their mothers, and had been in the social groups since birth (or in 3 cases since they were 2–3 weeks old). All mothers had been in the colony for at least a year before the experiments.

HOUSING

Each of 6 social groups occupied an outside cage (548 × 243 × 246 cm) communicating with an inside room (185 × 133 × 239 cm) (see Hinde & Rowell, 1962). During experiments infants and mother–infant dyads removed from the colony were housed in indoor cages (82 × 74 × 103 cm), 500 cm from the colony, in visual and tactile but not auditory isolation from other monkeys. Removed mothers were kept in rather smaller cages (60 × 60 × 80 cm), housed in remote huts out of sight and sound of other monkeys.

EXPERIMENTAL GROUPS

These are shown in Table 1. Separations were carried out when the infants were 30–32 weeks old.

DATA COLLECTION

The schedule is shown in Table 2. Watches were of the following types:

(*a*) *Infant's activity.* Infant's activity while off mother was recorded. Watch commenced at 10.00 h GMT or BST, and ceased when the infant

TABLE 2

SCHEDULE OF DATA COLLECTION

Groups	Pre-experimental	Pre-separation days			Separation/removal days						Post-reunion days								
	Infants' activity				4	5	9	12	15	16									
All		1	2	3							17	18	19	21		23	30	37	44
	Mother–infant				4	5	9	12	15	16									
M–I/R + (MI)/R	Maternal activity Social behaviour	Watches 1 month and 2 weeks before, and during pre-separation week									17 17	18		21	22	6 h of data		1 × 2 h watch	

Separation and/or removal occurred on Day 4, reunion on Day 17. Data were obtained on the days indicated. Data from the 3 pre-separation days were pooled. Data from the post-reunion days were pooled as indicated by the vertical discontinuous lines.

had been off the mother for 2 h, or at 14.30 h, whichever was the sooner.

(*b*) *Mother–infant data.* Data on interaction between mother and infant, collected simultaneously with (*a*).

(*c*) *Maternal activity* (M–I/R and (MI)/R groups only). Mothers' activity over a 2 h period from about 10.00 h.

(*d*) *Social behaviour* (M–I/R and (MI)/R groups only). Data on mothers' and infants' interactions with group companions over a 2 h period from about 10.00 h.

Methods of data collection were similar to those discussed in Hinde (1973, p. 398).

STATISTICAL ANALYSIS

Changes in measures between the experimental periods were assessed by Friedman two-way analyses of variance and, where significant, assessed further by Wilcoxon matched-pairs tests. Differences between groups were assessed by Mann-Whitney *U* tests (Siegel, 1956). All significance levels are two-tailed.

BEHAVIOURAL MEASURES

For clarity, not all the measures taken are discussed here. Those used have been described in the papers cited, and most of them are self-explanatory, but details of some are given below.

Activity (No. boxes). The outside pens were divided into 16 'boxes' (8 in plan, with upper and lower sections): activity was assessed in terms of the number occupied per half minute.

Activity when active ($> 5/ > 1\ \%$). The number of half minutes in which the infant occupied more than 5 box areas divided by the number of half minutes in which they entered more than one box, expressed as a percentage ($> 5/ > 1\ \%$).

Time spent sitting (S %). The number of half minutes in which the infant sat doing nothing as a percentage of the number of half minutes it spent off its mother.

Time off (T.T. off). The number of half minutes that the infant spent off its mother as a percentage of the total number of half minutes watched.

Relative number of rejections made by the mother to the infant's attempts to gain ventro-ventral contact ($R/Mk_I + Mk_M + R$), i.e. the number of times the infant attempted to gain ventro-ventral contact and was rejected by its mother (R), divided by the number of times the infant successfully gained v-v contact on its own initiative (Make Contact by Infant, or Mk_I), plus the number of times it gained v-v contact on the mother's initiative (Make Contact by Mother, or Mk_M), plus the number of times its attempts to gain v-v contact were rejected by the mother (R).

The infant's role in maintaining v-v contact ($\%Mk_I - \%Bk_I$). The number of 'makes', i.e. successful attempts to gain v-v contact by the infant (Mk_I), divided by the number of successful attempts to gain v-v contact initiated by the infant plus those initiated by the mother ($Mk_I + Mk_M$), minus the number of 'breaks', i.e. the number of times the infant broke v-v

contact (Bk_I), divided by the number of breaks by the infant plus the number of breaks by the mother ($Bk_I + Bk_M$), expressed as a percentage. The use of this index is discussed in Hinde & White (1974): a positive value indicates the infant was primarily responsible for contact.

Time spent at a distance (> 60 cm %). The number of half minutes in which the infant was more than 60 cm from its mother (i.e. over arm's reach), as a percentage of the number of half minutes it was off its mother.

The infant's role in maintaining proximity ($\%Ap_I - \%L_I$). The number of approaches made by the infant (i.e. coming within 60 cm of the mother, Ap_I) divided by the number of approaches made by the infant plus the number of approaches made by the mother to the infant ($Ap_I + Ap_M$), minus the number of leaves by the infant from its mother (L_I) divided by the number of leaves made by the infant plus the number of leaves made by the mother ($L_I + L_M$) expressed as a percentage. A positive score indicates that the infant was primarily responsible for proximity (see Hinde & Atkinson, 1970).

RESULTS

PRE-SEPARATION PERIOD

There were no significant differences in measures of the mother–infant relationship. There were, however, some differences in infant activity: infants in the group in which both mother and infant were removed and separated (M–I/R) were less active than the Mother-Removed (M/R) ($P < 0{\cdot}08$) and Infant-Removed (I/R) ($P < 0{\cdot}02$) infants, and less active when active than both M/R and I/R infants ($P < 0{\cdot}002$). Infants in the group in which both Mother and Infant were Removed but not Separated ((MI)/R) were less active when active than the M/R and I/R infants ($P < 0{\cdot}08$ and $P < 0{\cdot}006$). The (MI)/R infants, however, engaged in significantly more rough-and-tumble play than the I/R infants ($P < 0{\cdot}10$). Comparisons of mothers were possible only

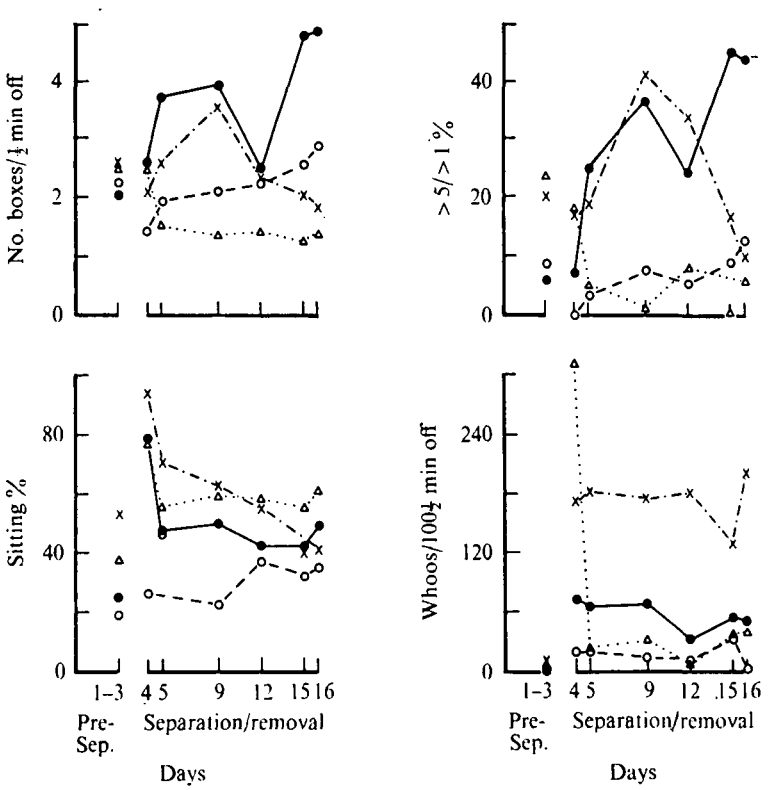


FIG. 1. Behaviour of infants before and during separation/ removal – group medians. O– – –O, (MI)/R; ×–·–·–×, I/R; ●—●, M–I/R; △....△, M/R.

between M–I/R and (MI)/R groups, as no data were available for the others. M–I/R mothers were less active when active ($P < 0{\cdot}06$) but spent more time being groomed than the (MI)/R mothers ($P < 0{\cdot}06$)

THE SEPARATION/REMOVAL PERIOD (Fig. 1)

The behaviour of all infants separated from their mothers could be adequately characterized as involving bouts of protest (high rate of calling, moderate or high locomotor activity) alternating with bouts of despair (moderate to low calling, moderate or low activity, often sitting inactive). The former were more common early in the separation period, the latter later. The period of frequent protest lasted longer in infants experiencing separation in a strange environment (I/R, M–I/R) than in those remaining in the home pen (M/R): the latter showed intense but relatively brief protests, depressive symptoms appearing after about 24 hours.

Throughout the separation/removal period the infants of the removed but non-separated group ((MI)/R) showed few changes from their pre-separation behaviour. There was a slight increase in whoo calling, but this was significant only on Day 9. On the day of separation all other groups showed significant ($P < 0{\cdot}10$) increases in whoo calling to levels significantly greater than those of the non-separated infants ($P < 0{\cdot}10$), but did not differ significantly from each other. Infants of groups in which the infants were removed and separated (I/R and M–I/R) retained their high levels of calling throughout the separation period: the difference between them, though considerable, was significant only on Day 12 ($P < 0{\cdot}02$). By contrast the Mother-Removed (M/R) infants, which had the highest median rate of calling on the day of separation, subsequently called little.

Infants of I/R, M–I/R and (MI)/R groups were kept in comparable cages during the separation/removal period, and their locomotor scores were comparable. Until Day 12 the non-separated group ((MI)/R) showed lower locomotor scores than either of the others, the difference from M–I/R continuing throughout the period (significant on Days 4, $P < 0{\cdot}10$; 5, $P < 0{\cdot}02$; 16, $P < 0{\cdot}10$). The non-separated group also scored less on sitting inactive than all other groups on each day (difference from I/R and M/R significant on Days 4, $P < 0{\cdot}002$; 12, $P < 0{\cdot}10$). Thus infants separated from their mothers call more, show more locomotor activity and sit inactive more, than infants remaining with their mothers, even though the latter are in a strange environment. The activity of these infants was in fact different in kind from that of the non-separated infants: they tended to alternate between periods of intense activity (high activity when active scores) and periods of sitting inactive.

Measures of locomotor activity were not comparable between the infants of these groups, who were placed in small cages, and the M/R infants, who remained in the large home pens. However, the trends through the separation period differed markedly between the separated groups. Whilst the M/R group showed higher medians for locomotor activity and for activity when active on the day of separation than on any subsequent day, the M–I/R and I/R groups were lower on the first day of separation than on most subsequent days.

Similarly, in frequency of sitting inactive the M/R group scored significantly less than the I/R group and was comparable to the M–I/R group on Day 4, but showed higher medians than either towards the end of the separation period, the difference from M–I/R being significant on Day 12 ($P < 0{\cdot}02$).

Thus the data show that the infants which remained in their home environment showed initially high levels of calling which quickly fell and initially moderate levels of locomotor activity which also fell, while infants moved into small cages maintained their elevated levels of calling, and maintained at least for several days (I/R) or throughout the separation period (M–I/R) high and even increasing overall levels of locomotor activity. The conditions during separation thus have a profound effect on the infant's behaviour.

Data on the mother's behaviour during the separation period were not complete. Mothers separated from their infants showed much locomotor activity and considerable whoo calling on Day 4. The (MI)/R mothers also showed some increase in locomotor activity and whoo calling on the day of removal, but significantly less than the mothers separated from their infants.

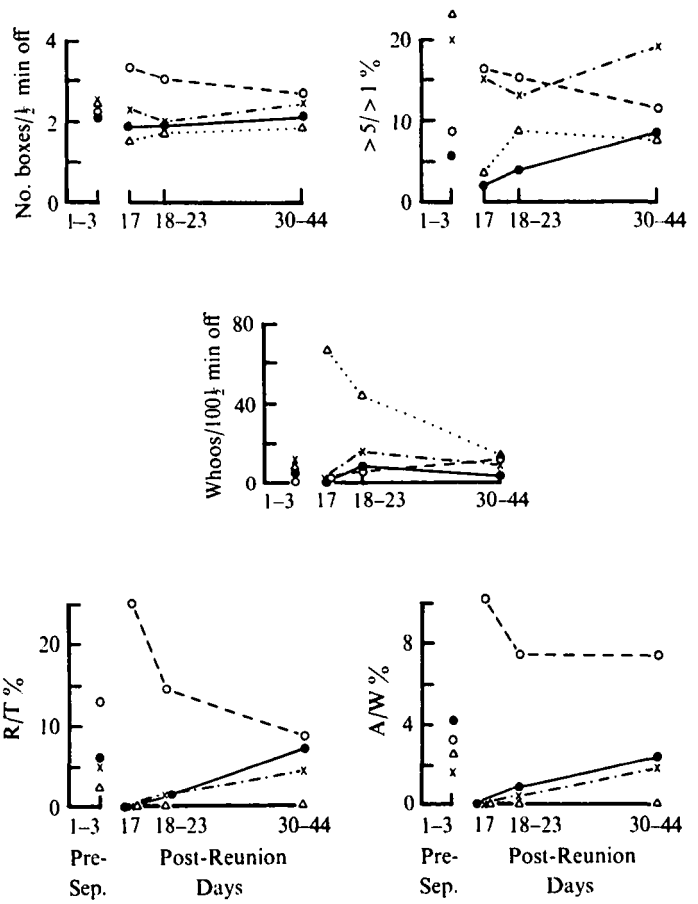


FIG. 2. Behaviour of infants before separation/removal and after reunion – group medians. O– – –O, (MI)/R; ×–.–.–×, I/R; ●—●, M–I/R; △..△, M/R.

In qualitative terms the changes subsided as the period progressed but quantitative data are not available.

POST-REUNION (Fig. 2)

(i) *Behaviour of the infants when off their mothers*
First those infants which had not been separated from their mothers ((MI)/R) differed from the other groups in that

(*a*) Their locomotor scores tended to be higher than before removal (no. boxes $P < 0{\cdot}04$, 0·04 and NS in successive periods; activity when active $P < 0{\cdot}04$, 0·04 and 0·10).

(*b*) Their locomotor scores tended to be higher than those of all other groups (no. boxes consistently higher, significantly so in 8 out of 9 comparisons; activity when active significantly higher in 6 out of 9 comparisons, but significantly lower than I/R in the last period).

(*c*) They played more than infants of other groups (score significantly higher in 7 out of 9 comparisons for rough-and-tumble play (R/T) and 8 out of 9 for approach/withdrawal play (A/W)), the differences being especially marked just after reunion.

(*d*) They continued to show only low levels of whoo calling.

In summary, these infants showed far fewer indications of distress than infants of the other groups, and their activity levels were high.

Of the separated groups, those in which the infants had been removed from the home pen (I/R and M–I/R) tended to have rather similar

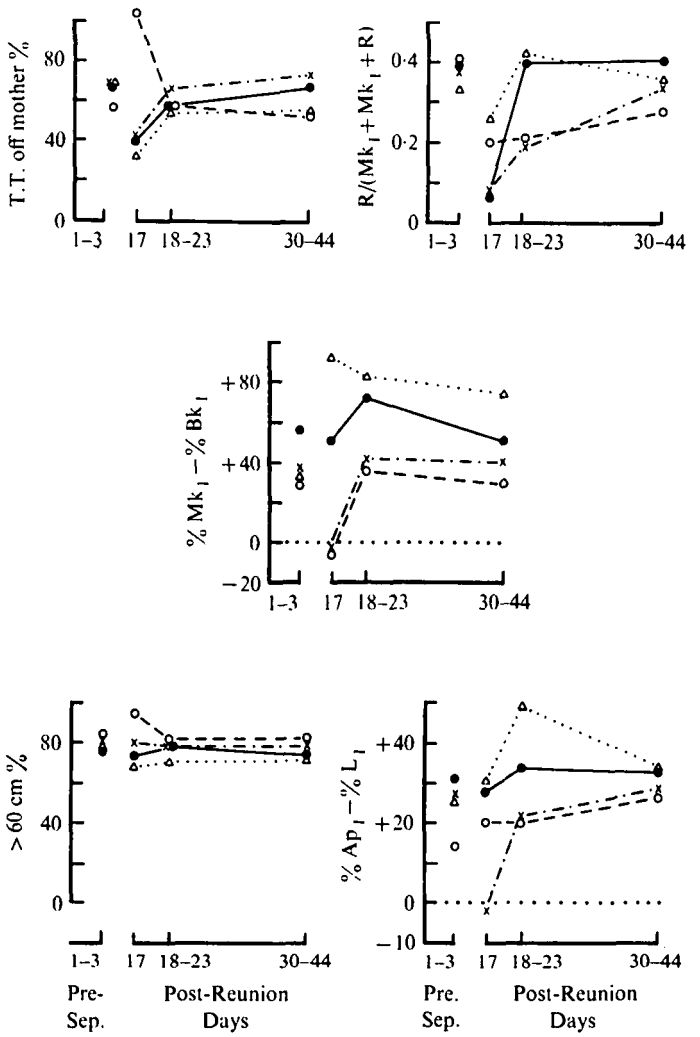


FIG. 3. Measures of mother–infant relationships before separation/removal and after reunion – group medians.
O---O, (MI)/R; ×-.-.-×, I/R; ●—●, M–I/R; △....△, M/R.

scores on all measures after reunion, except that the pre-separation differences in the 2 activity measures tended to persist (difference in activity significant in third post-reunion period, $P < 0{\cdot}02$, and difference in activity when active significant in the second, $P < 0{\cdot}02$, and third, $P < 0{\cdot}002$).

However, the separated infants which had remained in their home pen differed markedly from these 2 groups:

(*a*) The low locomotor scores they had shown during separation persisted, so that they continued to show lower levels of activity than before separation (difference in no. boxes $P < 0{\cdot}05$ and difference in activity when active $P < 0{\cdot}05$ in all 3 periods).

(*b*) Their locomotor scores thus tended to be lower than the other separated groups (no. boxes significantly lower than I/R in all 3 periods and than M–I/R in the third; activity when active significantly lower than I/R in the third period).

(*c*) Their median play scores were zero throughout the post-reunion period.

(*d*) Their calling rates were consistently higher than those of the other separated groups (both differences significant in first periods, $P < 0{\cdot}02$).

Thus (*a*) infants which had experienced

separation were more disturbed than those which had not, and (*b*) infants which during separation had remained in the home pen, and become depressed there, continued to be depressed and were more disturbed after reunion than infants of the other separated groups.

(ii) *Mother–infant relationship* (Fig. 3)

The differences between groups in the behaviour of the infants were closely related to differences in the mother–infant relationships.

The infants that had not been separated from their mothers gave low scores on $\%Mk_I - \%Bk_I$ (differences on Day 17 from M–I/R, $P < 0{\cdot}02$, from M/R, $P < 0{\cdot}001$; Days 18–23 from M–I/R, $P < 0{\cdot}10$ and M/R, $P < 0{\cdot}003$; Days 30–34 from M/R, $P < 0{\cdot}02$) and $\%Ap_I - \%L_I$. They thus did not show the strongly mother-directed behaviour shown by the separated infants. Their relative frequency of rejections was comparable to those of other groups on Day 17, but their time off mother was significantly higher ($P < 0{\cdot}02$ or less): since the time off increased from pre-separation values ($P < 0{\cdot}05$) whilst the relative frequency of rejections tended to decrease (NS), the change in time off from the pre-separation condition must have been due more to greater independence of the infants than to stronger tendencies of the mothers to reject the infants. Later the relative frequency of rejections tended to be lower than in the separated groups: differences in time off and in time at a distance from the mother were small. In comparison with the other groups, both mothers and infants were secure in their relationship with each other, and 'could afford' time for other social activities (see below) whilst maintaining frequent visual contact with each other.

Comparing now the 2 groups in which the infants were removed from the group and separated, the scores were generally similar. However, the M–I/R group tended to show a higher relative frequency of rejection (Days 18–23, $P < 0{\cdot}02$), and to score more highly on $\%Mk_I - \%Bk_I$ (NS) and $\%Ap_I - \%L_I$ (Day 17, $P < 0{\cdot}10$). Thus the similar times off and at a distance from the mother shown by these two groups involved more mother-directed behaviour from the M–I/R infants than from the I/R ones. The bases of these differences are discussed below.

Finally, the infants separated and left behind in their home pens tended to be rejected more than the other separated groups (Day 17, difference from M–I/R, $P < 0{\cdot}10$; Days 18–23, I/R, $P < 0{\cdot}02$), and scored more highly on $\%Mk_I - \%Bk_I$ (Day 17, differences from I/R and M–I/R, $P < 0{\cdot}10$; Days 18–23, I/R, $P < 0{\cdot}02$) and on $\%Ap_I - \%L_I$ (NS). This group spent less time off their mothers (Days 30–44, difference from I/R, $P < 0{\cdot}10$) and less of their time off at a distance from them (Days 30–44, differences from I/R and M–I/R, $P < 0{\cdot}02$ and $0{\cdot}10$) in spite of the fact that the infants had to attempt to get on their mothers more than the infants of the other separated groups.

In summary, the times the infants spent off and at a distance from their mothers differed little between groups, at any rate after Day 17. But the non-separated infants obtained their time on or close to their mothers with less striving than the separated infants. Amongst the latter, the M/R infants had to show more mother-directed behaviour to maintain their contact and proximity than did the infants of the other separated groups. Their high scores on the infant's role in contact and proximity occurred in spite of their low levels of activity, and imply that their mothers were very unresponsive. The differences between this group and that in which mothers and infants had been removed from the home pen and separated were probably due to the continuing depression of the infants that had been left in their home pen. The differences between the M–I/R and I/R groups were probably related to the facts that the mothers of the former had to re-establish their relationships with their group companions, whilst those of the latter had been in the home pen throughout. This is discussed in the next section.

RELATIONSHIPS WITH OTHER SOCIAL COMPANIONS

(Some measures shown in Fig. 4)

In the early experiments in this series, all separations followed the M/R procedure: in this way the infant was left in the physical surroundings and with the social companions it had known since birth. Later I/R experiments were added in the expectation that the infants, experiencing the trauma of strange surroundings in addition to

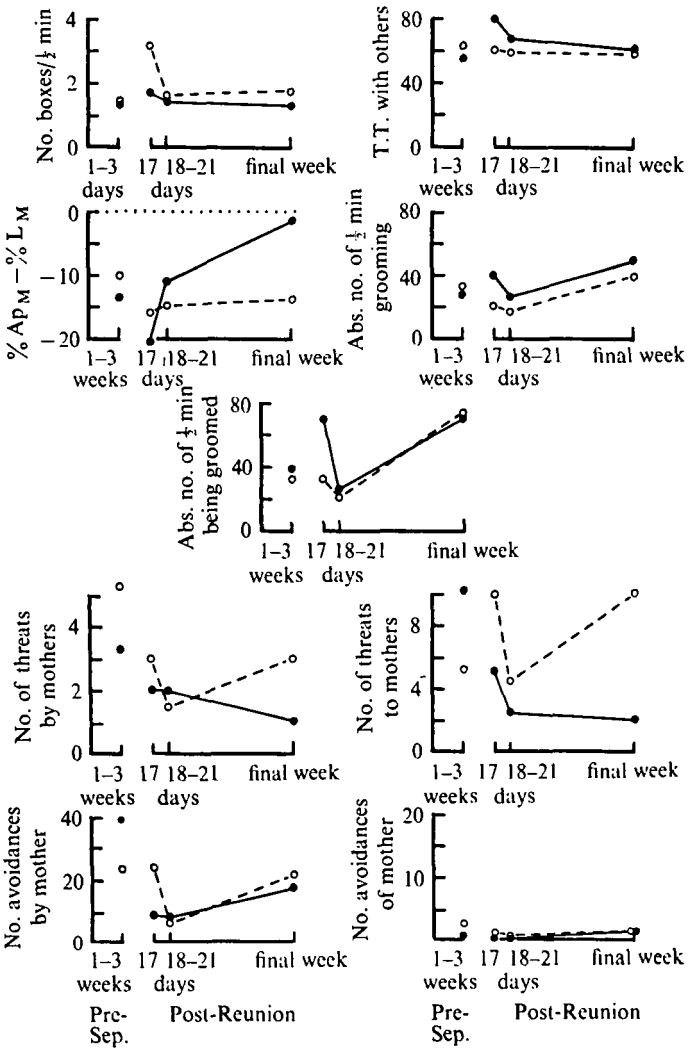


FIG. 4. Mothers' social interactions with group companions before separation/removal and after reunion – group medians for M–I/R and (MI)/R. O- - -O, (MI)/R; ●—●, M–I/R.

separation from the mother, would be even more distressed after reunion. The opposite proved to be the case, and it became apparent that the greater distress of the M/R infants after reunion was related to greater disturbance to their mother–infant relationships: in brief, the mothers, on return to the home pen, had to re-establish relationships with their other social companions, and were less tolerant of the demands of their infants (Hinde & Davies, 1972*b*).

Unfortunately, in those experiments no detailed data on the interactions with the other social companions were collected. The only comparisons possible, therefore, are between the M–I/R and (MI)/R groups.

On return to the home pen, the M–I/R mothers were re-united with their infants whilst the (MI)/R mothers had just come from a period of close confinement with theirs. In addition, however, some pre-separation differences must be taken into account – the M–I/R mothers tended to have somewhat lower social status in their respective groups, and spent less time sitting in contact with others ($P < 0{\cdot}10$). On reunion

with their groups, 1 of the M–I/R mothers rose in rank, whilst 2 of the (MI)/R mothers fell: the latter changes were foreshadowed by an instability in social relationships in the pre-separation period.

On Day 17 the M–I/R mothers divided their attentions between their infants and other social companions. Thus the (MI)/R mothers played the more active role in re-establishing their social relationships with group companions. Perhaps as a result, the (MI)/R mothers not only threatened others more and were avoided by others more than the M–I/R mothers, a continuation of the pre-separation trends, but also were threatened by them and avoided them more, a reversal of pre-separation trends (difference NS). Furthermore, the (MI)/R mothers spent less time in contact ($P < 0{\cdot}10$) and with ($P < 0{\cdot}10$) others and more time alone ($P < 0{\cdot}002$) than did the M–I/R mothers, again a reversal of pre-separation trends.

On the day of reunion, the infants that had been separated showed more mother-directed behaviour (see above) and tended to avoid other social companions: they spent less time in social, locomotor and manipulative play (each $P < 0{\cdot}002$) than did the (MI)/R infants. Like their mothers, they threatened and were threatened by, avoided and were avoided by, others less than the (MI)/R infants ($P < 0{\cdot}10$, $0{\cdot}002$, $0{\cdot}10$, $0{\cdot}02$ respectively). The infants who had not been separated, although off their mothers more than the M–I/R infants, continued to make frequent visual reference to them while off them.

After the day of reunion, there were no further significant differences in measures of the mothers' relationships with their group companions. Differences between infants in the amount of social and locomotor play continued (Days 18–21, NS and $P < 0{\cdot}02$; Days 30–44, NS): that in manipulative play, still present on Days 18–21 ($P < 0{\cdot}02$), was reversed in the last week ($P < 0{\cdot}01$).

The most significant issues emerging from this comparison were the marked differences between the groups on the day of reunion. In the M–I/R group, the infants directed most of their behaviour towards their mothers, and the mothers divided their attentions between infants and social companions, whereas in the (MI)/R group both mothers and infants directed a high proportion of their attention to social companions. While most of the differences ceased to be significant after the first day, many of the trends continued, and the M–I/R infants continued to show more mother-directed behaviour and less social play throughout.

Perhaps even more revealing were the individual differences within groups. In the M–I/R group the lower-ranking mothers appeared especially to avoid interactions with group companions and, perhaps because of this, were also more rejecting towards their infants. Their infants were therefore more distressed throughout the reunion period. In the (MI)/R group the lower ranking mothers tended at first to be more active, and their infants had to play a greater role in maintaining proximity than did other infants, and interacted less with others: at this stage these were the more distressed infants. Later the more distressed infants were those whose mothers had unstable relations, and especially the infants of the 2 females which fell in rank.

These data are in harmony with the view that mothers removed from their social group must re-establish their social relationships on return, and that this may affect their relationships with their infants (and vice versa). It also suggests that the magnitude of this effect depends on the nature of the mothers' pre-existing social relationships.

DISCUSSION

These experiments clearly demonstrate the multiplicity of factors that influence the consequences of a separation experience. We may now consider briefly the interacting factors demonstrated here and in other experiments on non-human primates.

(1) SEPARATION

Infants that had been separated from their mothers exhibited more distress than infants who had merely been removed from their social group with their mothers. Thus separation and its concomitants can be said to have an effect. This, of course, is not to say *how* separation produces its effect. Two possibilities are that it is the accompanying 'emotional deprivation' (e.g. Howells, 1970) and/or 'stress' (Robertson & Robertson, 1971) that are important. Another, not incompatible with these, is that a separation experience

produces its effects in part by influencing the course of the mother–infant relationship: some of the factors discussed below as influencing the outcome of a separation experience may act in this way. In general, it may be that the several concomitants of a separation experience may produce both distinguishable and interacting consequences on the developing infant (cf. Hofer, 1972).

(II) THE PRE-SEPARATION MOTHER–INFANT RELATIONSHIP

In a previous series of experiments involving 6 day M/R separations, the infants more distressed in the post-reunion period were those which, before separation, had had the highest proportion of their attempted nipple contacts rejected, and had had to play the greatest relative role in staying near their mothers when off them. How much time the infants had spent on or near their mothers was not correlated with post-reunion distress: it thus seemed that, speaking colloquially, 'tension' in the mother–infant relationship was more relevant than the actual amount of time spent together (Hinde & Spencer-Booth, 1971; Spencer-Booth & Hinde, 1971*b*). Comparable data were obtained in the M–I/R and (MI)/R groups.

There are parallels here with studies of human children, who likewise appear to be less affected by a separation experience if they had good relationships with their mother before separation (Vernon *et al.* 1965).

(III) POST-REUNION MOTHER–INFANT RELATIONSHIPS

In the experiments with rhesus monkeys referred to above, the values of the 'tension' measures were correlated between pre-separation and reunion periods: the correlations between pre-separation relationships and post-reunion distress could thus have been mediated by differences in the post-reunion relationship. Examination of partial correlation coefficients showed that on the day of reunion infant distress was related more closely to the pre-separation than to the contemporaneous measures of 'tension', but subsequently this situation was reversed (Hinde & Spencer-Booth, 1971). This suggests that infant distress immediately after reunion is determined more by the pre-separation relationship than by the contemporaneous one, but after a day or two the latter becomes prepotent.

The importance of the post-separation mother–infant relationship is confirmed by the inter-group differences in the current experiments. Furthermore, within groups the infants with the most tense relationships with their mothers tended to be the more disturbed.

The post-reunion relationship may be affected also by the infant's behaviour after reunion. Human children (e.g. Bowlby, 1973), and less often monkeys (Rosenblum, 1971), often show a period of 'detachment' from their mothers after reunion. Human children may also show hostility (Robertson & Robertson, 1971). If mishandled by the parent, such changes could lead to a permanent deviation in the relationship. In the case of one girl the Robertsons are inclined to ascribe a discontinuity in the relationship resulting from a separation experience to a 'subtle disturbance of her super ego development': no doubt experiences of either mother or child during the separation period could affect their subsequent relationship. In any case, it cannot be doubted that the subsequent course of that relationship is of crucial importance.

(IV) PRE- AND POST-SEPARATION MATERNAL SOCIAL RELATIONSHIPS

The mother–infant relationship both before and after separation is affected by the nexus of other social relationships in which it is embedded (Hinde & Spencer-Booth, 1967). The mothers in the M–I/R and (MI)/R groups with the most tensionful pre-separation relationships tended to be those with the lower ranks in their social groups. Such mothers tended to be more active in avoiding social interactions with their group companions, and to provide a less adequate 'secure base'. There were indications that it was not only low rank, but inconsistent social relationships in the mothers that was associated with tense mother–infant relationships. Comparable data are available in the human case (e.g. Robertson, 1965). Furthermore, other data from this colony indicate that the mother–infant relationship varies with a variety of other factors,

including the previous experience of the mother, the sex of the infant and the presence of peers (White & Hinde, 1975; see also Kaplan, 1972; Rosenblum, 1974). The rearing environment also may be an important variable (Castell & Wilson, 1971; Brandt *et al.* 1972).

Evidence that the effects of mother–infant separation in man may be markedly affected by the mother's other social relationships, especially that with her spouse, has been reviewed by Rutter (1971).

(v) PRE-SEPARATION INFANT SOCIAL RELATIONSHIPS

In both man (e.g. Mead, 1962) and monkey, the effects of separation are also likely to be less if the infant before separation had social relationships with other individuals in addition to their mothers. Thus Kaufman & Rosenblum (1969) found that bonnet macaque infants had less exclusive relationships with their mothers and related to others more readily in their mothers' absence than did pigtail macaque infants (see (viii) below).

(vi) NATURE OF SEPARATION

During the separation period itself, the infants in the M/R group showed a more rapid onset of symptoms of depression than did those in the I/R and M–I/R groups. In terms of the natural situation, this might make adaptive sense: infants separated from their mothers within the group might do better if they conserve their energies until their mothers find them, whilst infants which are separated from the troop must seek for it and make their presence known (Kaufman, 1974). The mechanism, however, is unknown.

The depression tended to persist after reunion, and appeared to have long-term effects on the mother–infant relationship. This persistence could be a consequence of adverse conditioning to the home environment during the separation period (Gewirtz, 1961). This may have been exacerbated by the fact that M/R infants, although they showed high levels of distress calling during the reunion period, continued to show depressed locomotor activity and thus did not keep so close to their mothers as did the I/R and M–I/R infants.

The effects of separations induced by removing infant or by removing mother rhesus monkeys to a strange pen, have been compared also by Chappell & Meier (1975). Their procedure, however, differed in important ways from that used here. Each of the 4 mothers lived in a small cage with a small exit into a common play area: the exit permitted the infants to come and go, but was too small for the mothers. Thus the mothers had no access to each other, and the question of re-establishing relationships with social companions after reunion could not have arisen. Furthermore, separations lasted 6 days and were repetitive. The treatment differences during the separation period were comparable to those found in the present experiments: mother-removed infants showed symptoms of depression, whilst infants that were themselves removed to a strange pen become hyperactive. However, after reunion the results differed from those of the experiments reported here: infant-removal caused a larger increase in ventro-ventral and nipple contact than mother-removal. Interestingly, Chappell & Meier found that after infant-removal infants clung significantly more often than mothers cradled, but this was not the case after mother-removal. In view of the procedural differences (e.g. shorter and repeated separations, different social situations of mothers) it is not profitable to discuss the apparent discrepancy between these experiments and our own in the relative effects of the 2 treatments after reunion: obviously several explanations are possible.

In the human case it seems likely that the stress of separation is greater if the infant is moved to strange surroundings than if the infant remains at home (Douglas & Blomfield, 1958; Rutter, 1972). The significant finding from the animal experiments, however, would seem to be that the immediate effects are *different*, protest lasting for longer and depression being delayed if the infant is in strange surroundings, than if it remains in the familiar physical social situation. With both human (Rheingold, 1969; Fagin, 1966) and non-human infants, distress in strange surroundings is ameliorated by the presence of the mother.

(VII) DURATION OF SEPARATION

In rhesus monkeys the effects of one 13-day M/R separation experience are greater than those of two 6-day experiences, and these in turn are greater than those of one 6-day experience (Spencer-Booth & Hinde, 1971*b*). A similar conclusion is probable also in the human case (refs. in Rutter, 1972).

(VIII) ALTERNATIVE SOURCES OF MOTHERING DURING THE SEPARATION PERIOD

Infants subjected to the M/R procedure were left with social companions during the separation period, and were able sometimes to snuggle up to them. Indeed a few infants were held ventro-ventrally and carried by adult females other than their mothers. However, there were no clear differences during the reunion period between infants who had and had not had contact with adults other than their mother during the separation period. Schlottman & Seay (1972) found that the presence of an unfamiliar adult female at most slightly accelerated the recovery during a 3-week separation experience of young Java monkeys (*Macaca irus*) after reunion with their mothers. However, in these species adult females are not so attracted to young infants as are the females of some other species: Kaufman & Rosenblum (1969) found that bonnet infants whose mothers had been removed were less distressed than pigtail infants, and related this to the fact that bonnet adults were more attentive to the infants than were the pigtail adults. That the effects of a separation experience on human children can be ameliorated or eliminated by sensitive fostering is well established (e.g. Robertson & Robertson, 1971). Indeed the Robertsons mention one case in which a well-managed supportive separation appears to have brought some benefits, enabling the child to initiate a richer interaction with her mother: such cases are probably rare.

(IX) SEX OF INFANT

In the earlier 6-day M/R experiments, there were some indications that males were upset more than females in the post-reunion period. In the present experiments the separated males tended to be more active but to whoo less than the females. It is difficult to know how much emphasis to place on such sex differences, or to know how much they could be ascribed to differences between mother–male infant and mother–female infant relationships. However, there is other evidence from non-human primates (Sackett, 1968) and man (Rutter, 1972) that males are more susceptible to social stress than females.

(X) AGE AT SEPARATION

Although this is undoubtedly an important variable, there have been few systematic studies. In rhesus monkeys, no systematic differences were found between the effects of 6-day separations at 21, 25–26 and 30–32 weeks, though separations at 18 weeks seemed to produce more severe effects (Spencer-Booth & Hinde, 1971*a*). Suomi *et al.* (1973) found that infants removed from their mothers at 90 days showed a more severe immediate reaction than infants separated at 60 or 120 days, but by the time the infants were 6 months of age the differences had largely disappeared. Squirrel monkeys separated from their mothers when they had reached a stage at which they were spending only 10% of the time on them showed relatively little disturbance (Kaplan, 1970).

(XI) OTHER CHARACTERISTICS OF THE INFANT

There are so far no data from non-human primates relating individual characteristics of the infant to response to separation. It seems certain, however, that such relations could be found. Robertson & Robertson (1971) mention 'ego immaturity' as one source of individual difference in man.

(XII) DIRECT EFFECTS OF SEPARATION ON THE MOTHER

These experiments were conducted when the infants were getting little milk from their mothers. However, 2 of the mothers from the M–I/R group were seen to have very slightly swollen and blistered nipples during the separation period.

Perhaps for this reason, their infants had to play high relative roles in maintaining contact and proximity after reunion. However, these mothers were also low-ranking, so the condition of their nipples may not have been an issue.

(XIII) MOTHERS' PREVIOUS EXPERIENCE

This was not controlled in the present study, but Kaplan (1970) found in squirrel monkeys that experienced mothers and their infants showed less 'attachment' than inexperienced mothers and their infants. Such conclusions are likely to depend crucially on the measure of 'attachment' used.

(XIV) INFANTS' PREVIOUS HISTORY OF SEPARATION

In experiments from this laboratory some infants were separated (M/R, 6 days) first when either 21 or 25–26 weeks old, and then again when 30–32 weeks old. Measures of infant behaviour after reunion from the second separation experience were compared with measures from infants given their first separation experience at 30–32 weeks. There were few significant differences, but some indications that the infants that had been separated for the second time were slightly less disturbed than those recently reunited from their first separation experience (see also Chappell & Meier, 1975). However, the long-term effects of 2 separation experiences were greater than those of one (Spencer-Booth & Hinde, 1971*a*). Data on repetitive separations of infant rhesus monkeys from their peers indicate a cumulative effect of the separations (Suomi *et al.* 1970; see also Young *et al.* 1973): here, however, the situation was different for the reunions after separation occurred with a similarly affected peer.

These data, like those of Robertson & Robertson (1971) and Suomi (1974), indicate that the consequences of a separation experience depend on diverse factors which interact in a complex way (see also Rutter, 1972): among such factors the present data emphasize especially the possible effects of the separation on the mother–infant relationship. Since certain parallels between findings from experiments with non-human primates and data on human children have been mentioned, it is perhaps as well to emphasize that a comparative approach must be used with circumspection. The importance of particular independent variables is likely to differ even between non-human primate species (Rosenblum & Kaufman, 1968; Preston *et al.* 1970; Kaplan, 1970; Schlottman & Seay, 1972), let alone between monkey and man (e.g. Lehrman, 1974; Hinde, 1976). In making comparisons, principles established from the data are likely to have greater generality than particular research findings. For example, it seems likely that human children are more likely to be upset by a separation experience if it involves strange surroundings (e.g. hospitalization) than if they stay at home. This differs from the findings reported here, but the principle that the effects of a separation experience vary with the degree of disruption of the mother–infant relationship is likely to be common to both. Language and related cognitive abilities may provide human relationships with techniques for withstanding temporary separations not available to monkeys.

Data from monkeys may also be relevant to the broader issue of the nature of the dynamics of relationships. Again because of their relative simplicity, they permit the isolation of principles which might be obscured by the complexity of the human case. In the present instance, separation experiments suggest that it may be profitable to consider the relationships as depending in part on negative feedback: after separation the infant's behaviour is more filial, but if the mother becomes more maternal the infant's behaviour gradually returns to normal. However, the infant's depression and the mother's preoccupations may exceed the capacities of the system, so that a long-term change in the relationship is produced (see Hinde & Stevenson-Hinde, 1976).

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